

DAY-11(30.07.2026)

▼ VPN VIRTUAL PRIVATE NETWORK:

- It is a server that connects device over the internet that access the web page by the user encrypts (secures the IP address of the user) data through virtual tunnel and the process is known as tunneling
- Normally when the internet is accessed through ISP which can see the log of the user but they are not secure which can be shared to third parties and it is not private anymore
- but with VPN , the third parties cannot be able to see what the user using or watching and it is more secure that the IP address is hidden by VPN to internet services

BENEFITS:

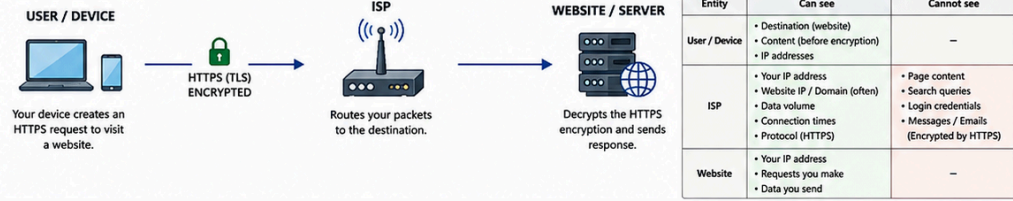
1. Privacy protection → encrypts data and IP address is hidden that third parties can't able to monitor browsing of the user
2. Security on public Wi-Fi
3. Bypass geo-restrictions
4. reduces slowing down of the internet services through ISP (ISP throttling)
5. Secure connection for remote access → employees to connect to companies from outside of organization (instead Leased line, VPN can be used which acts as secured private network which makes one private network virtually)

VPN, HTTPS & ISP – COMPLETE PACKET FLOW

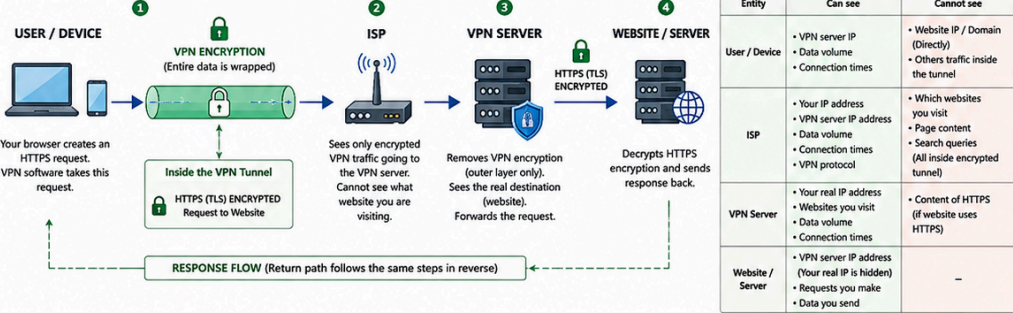
User → ISP → VPN → Website

■ Can see
 ■ Cannot see
 ■ Partial / Metadata only
 ■ Not applicable

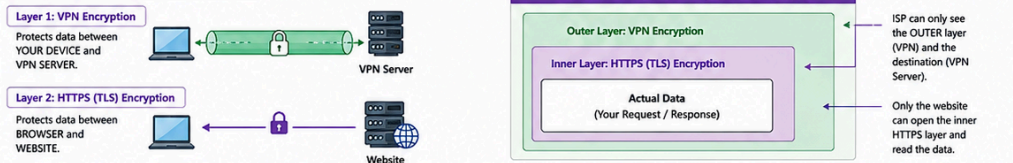
1) WITHOUT VPN



2) WITH VPN



3) ENCRYPTION LAYERS EXPLAINED



4) QUICK COMPARISON

Scenario	What ISP sees	What Website sees	Your real IP visible to Website?	Can ISP see which websites you visit?	Can ISP see what you do inside websites?
Without VPN (HTTPS only)	Your IP, Website IP/Domain (often), Data volume, Time, Protocol	Your real IP, Requests, Data you send	YES	Often YES (to a significant extent)	NO (HTTPS encrypted)
With VPN (HTTPS + VPN)	Your IP, VPN Server IP, Data volume, Time, VPN Protocol	VPN Server IP, Requests, Data you send	NO (Website sees VPN IP)	NO (Not easily / not normally)	NO (HTTPS encrypted)

KEY TAKEAWAYS

- ISP is required for Internet access. VPN works on top of ISP.
- HTTPS protects content. VPN protects your privacy from ISP (hides destination & IP from websites).
- VPN does not make you 100% anonymous. The VPN provider can see your activity (depending on its policy).
- Always choose a trustworthy VPN with a strict no-logs policy.

MEMORY RULE

HTTPS = Protects WHAT you say
 VPN = Protects WHERE you go (from ISP)

